

PATHCARE LABS

Enterprise SaaS Platform

MASTER EXECUTION BLUEPRINT

Architecture · Development · Business Analysis · Marketing & Sales

Modules	Screens	Timeline	Target ARR
11 Domains	47+ Screens	14 Months	₹60 Cr+

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PART A

Architecture & Technical Blueprint

PathCare Labs SaaS is a cloud-native, multi-tenant diagnostic lab management platform reverse-engineered from Advik Path Lab (Windows .NET desktop) and rebuilt as a modern React + Node.js SaaS product targeting Indian diagnostic labs — Tier-2/3 cities, multi-branch chains, hospital-attached labs, and radiology centres.

What We Are Building

- Multi-tenant cloud SaaS for diagnostic labs across India
- Target: Standalone labs, multi-branch chains, hospital labs, radiology centres
- Compliance: NABH · NABL · PC-PNDT · DPDP Act 2023
- Stack: Next.js 14 + NestJS + PostgreSQL + Redis + AWS

Business Impact Projection

Scale	Labs	Monthly Patients	Infra Cost/mo	Target ARR
MVP	10	~5,000	~\$300	~₹12L
Growth	100	~50,000	~\$1,500	~₹1.2 Cr
Scale	1,000	~5,00,000	~\$8,000	~₹12 Cr
Enterprise	5,000	~25,00,000	~\$25,000	~₹60 Cr

Strategic Transformation

FROM (Legacy .NET Desktop)	TO (Cloud SaaS)
Single lab, single PC	100s of labs, global access
Manual backup	Auto-backup, DR-ready
No mobile access	PWA + Mobile First
Static reports	Real-time dashboards + AI
No integrations	HL7/FHIR, ABHA, SMS, WhatsApp
Zero scalability	Kubernetes, auto-scale

11 Domains · 47 Screens

Module Criticality & Priority Matrix

Module	Criticality	User Role	Frequency	Phase
Booking (Patient Reg + Test Order)	 Critical	Receptionist	50–200x/day	P1
Test Result Entry	 Critical	Lab Technician	50–200x/day	P1
Workload Dashboard	 Critical	Supervisor	Continuous	P1
Balance Receipt	 Critical	Cashier	20–100x/day	P1
Patient Registration	 Critical	Receptionist	30–150x/day	P1
Test Master / Doctor Master	 High	Admin	Setup+weekly	P1
MIS Panel (20+ Reports)	 High	Manager/Owner	Daily	P1
Internet Reporting (Email)	 High	Lab Tech	Daily	P1
Radiology Result Report	 Medium	Radiologist	10–30x/day	P2
Doctor Billing / Commission	 Medium	Accounts	Weekly/Monthly	P2
CMO Entry + PC-PNDT Register	 Medium	Radiology	Per USG scan	P2
Out Source Lab Management	 Medium	Admin/Tech	As needed	P2
Barcode Master	 Medium	Lab	Daily	P2
Daily Voucher / Petty Cash	 Standard	Accounts	Daily	P3
Delete Booking (Admin only)	 Admin	Owner only	Rarely	P3

Modular Monolith → Decomposable Microservices

Architecture Philosophy

Start as a Modular Monolith to avoid microservices operational complexity on Day 1. Build clean domain boundaries (NestJS modules) that allow individual service extraction when scale demands. This gives startup speed with enterprise upgrade path.

Multi-Tenancy Strategy: Schema-per-Tenant (PostgreSQL)

Strategy	Data Isolation	Cost	Complexity	Decision
Shared schema (row-level)	Low	Lowest	Low	✗ Rejected
Schema per tenant ✓	High	Medium	Medium	✓ CHOSEN
DB per tenant	Very High	Highest	High	Enterprise only

Technology Stack Summary

Layer	Choice	Why	Alternative
Frontend Framework	Next.js 14 (App Router)	SSR+SSG+RSC, SEO, file routing	Vite + React
Language	TypeScript 5	Type safety, enterprise grade	JavaScript
UI Library	shadcn/ui + Tailwind	Headless, accessible, customizable	MUI, Ant Design
Backend	NestJS 10 (Node.js)	Modular DI, TypeScript, RBAC guards	Express, Fastify
Primary DB	PostgreSQL 16 (RDS)	ACID, schema-per-tenant, RLS	MySQL, MongoDB
Cache/Queue	Redis 7 (ElastiCache)	Sessions, hot data, BullMQ queue	Memcached, RabbitMQ
Object Store	AWS S3 / MinIO	Reports, uploads, backups	GCS, Azure Blob
Cloud	AWS (Mumbai + Hyderabad)	India regions, BAA available, SES	GCP, Azure
Containers	Docker → ECS Fargate → EKS	Managed, progressive scaling	Bare metal, Heroku
SMS (India)	MSG91 (DLT registered)	India-first, DLT compliant	Twilio
Email	AWS SES	Cost-effective, DMARC support	SendGrid
WhatsApp	Twilio WhatsApp Business API	India primary communication channel	Gupshup

Layer	Choice	Why	Alternative
Monitoring	Grafana + Prometheus + Sentry	Full observability stack	Datadog
Testing	Vitest + Playwright + Jest	Fast unit + E2E + API tests	Jest + Cypress

Regulatory Framework

India diagnostic lab context requires alignment with:

- IT Act 2000 — India data protection baseline
- NABH Standards — accreditation requirements
- NABL ISO 15189 — lab quality standards
- PC-PNDT Act — prenatal diagnostic compliance (CMO register mandatory)
- DPDP Act 2023 — Digital Personal Data Protection (India's GDPR equivalent)
- HIPAA-aligned practices — for international/corporate clients

Security Layers

Layer	Control	Implementation
Transport	TLS 1.3 only	AWS ACM + Force HTTPS
Auth	JWT + MFA (TOTP)	Argon2id hashing, 5 failed = 15min lockout
Authorization	RBAC Guards	NestJS Guards + Permission Matrix per role
Database	Schema isolation + RLS	No cross-tenant leaks possible via SQL
Encryption	AES-256 at rest	RDS encryption + S3 SSE + pgcrypto for PHI fields
Aadhaar/PAN	Column-level masking	Always show last 4 digits only, never full
Audit	Immutable append-only log	Partitioned audit_logs, cryptographic chaining (P3)
Device Security	Device fingerprinting	New device login triggers extra verification
Reports	Dynamic watermarking	Patient name + date embedded in every PDF
DPDP Act 2023	Consent management	Data deletion requests, breach notification <72hrs
API Security	Rate limiting + WAF	Kong Gateway + Redis counters + AWS WAF

Performance Targets

Scenario	Target	Strategy
Booking save	< 300ms	Optimistic UI + indexed queries
Workload load (100 records)	< 500ms	Virtual scroll + pagination + partial index
MIS Day Collection report	< 2s	Read replica + materialized views
Test catalog autocomplete	< 100ms	Redis cache (1hr TTL)
Report PDF generation	< 5s	BullMQ async queue + Web Worker
Page initial load	< 1.5s	Next.js RSC + CDN + code splitting
API P99 latency	< 1s	PgBouncer connection pool + Redis cache

Database Optimization Indexes

Critical indexes required from Day 1 for production performance:

- idx_bookings_date_centre — booking_date + centre_id (WHERE status=Active) → Workload screen
- idx_bookings_patient — patient_id + booking_date DESC → Patient history
- idx_workload_pending — partial index on booking_tests WHERE result_status=Pending
- idx_patient_search — GIN index on to_tsvector for full-text name search
- idx_audit_entity — entity_type + entity_id + created_at for audit queries

PART B

UI/UX & Product Improvements

Biggest weakness in original document — now addressed



The original document selected shadcn + Tailwind but had zero design system strategy, no keyboard workflow design, and no offline-first plan.

Design System

- Design Tokens — spacing, color, typography all defined in one place. Every screen consistent.
- shadcn/ui + Tailwind CSS — headless, fully accessible, keyboard-navigable out of the box.
- Dark mode — lab staff stare at screens 8+ hours. Eye strain = errors. Dark mode is mandatory.
- Skeleton loaders — on slow Indian broadband, loading spinners kill UX. Skeleton UI = perceived speed.
- Storybook — every UI component documented and tested in isolation before integration.

Keyboard-First UX (CRITICAL for Receptionists)



A receptionist doing 200 bookings/day cannot use a mouse for every action. Keyboard shortcuts = 3x faster workflow = competitive advantage no competitor has.

Shortcut	Action	Screen
F1	New Booking	Global
F2	Result Entry	Global
F3	Balance Receipt	Global
F4	Workload Dashboard	Global
F5	Patient Search	Global
Ctrl+S	Save current form	All forms
Ctrl+P	Print report/receipt	Report screens
Escape	Cancel/Close modal	All modals
Tab / Shift+Tab	Navigate form fields	All forms
Ctrl+B	New booking for same patient	Booking screen

Offline-First PWA (CRITICAL for Indian Labs)

**RISK**

Indian Tier-2/3 city labs have frequent internet outages. If software stops working = lab stops earning. Offline-first is a must-have, not nice-to-have.

- Service Worker — cache critical screens (booking form, test catalog, patient list) in browser
- IndexedDB — store bookings locally when offline, sync when connection restores
- Sync queue — queued operations shown to user, auto-retry on reconnect
- Conflict resolution — server wins for most data, UI shows merge conflicts
- Offline indicator — clear banner showing "Working Offline — Data will sync when connected"

Mobile Responsive Design

- Collection boys in field — need mobile-friendly sample collection list
- Lab owner — needs mobile dashboard to check daily collection while travelling
- Radiologist — may want to review and approve reports from tablet
- Breakpoints: Mobile (< 640px), Tablet (640–1024px), Desktop (> 1024px)

Missing from original document — now added

WebSocket Integration (Socket.io)

- Live workload updates — when technician enters result, receptionist sees status change instantly without refresh
- Live TAT timer — workload screen shows real-time countdown for each pending test
- Multi-user conflict detection — if 2 technicians open same result entry simultaneously, show warning
- Live daily collection counter — owner dashboard auto-increments as payments come in
- Real-time SMS/WhatsApp delivery status — show tick marks like WhatsApp (sent → delivered → read)

Event-Driven Architecture

Event	Trigger	Action
booking.created	New booking saved	SMS to patient, workload update, barcode print
result.entered	Technician saves result	Workload refresh, TAT calculation
result.completed	All tests of booking done	WhatsApp/SMS/Email report delivery
payment.received	Balance receipt created	Live collection counter update
tat.breached	TAT SLA exceeded	Alert to supervisor, PagerDuty
result.critical	Critical value flagged	Urgent SMS to patient + doctor

Missing Features in Original Document

Feature	Description	Impact	Phase
TAT SLA Alerts	If CBC > 4hrs pending, supervisor gets alert	● High	P1
Auto-retest Flagging	Extreme result → suggest retest automatically	● High	P1
QR on Reports	Patient scans QR, verifies report online (anti-forgery)	● High	P1
WhatsApp Delivery	Report + status via WhatsApp Business API	● High	P1
Package Discounts	3+ tests together = auto% discount (configurable)	● Med	P1
Bulk Test Import CSV	Import 500 tests from CSV — critical for onboarding	● High	P1
Corporate Billing	Company employees → monthly invoice separately	● Med	P2
Insurance/TPA Billing	Cashless patient billing via TPA	● Med	P2
Inventory (Basic)	Track reagent stock — tests remaining before reorder	● Low	P3
Report Versioning	Correction audit trail — old version preserved	● Med	P2
Dynamic Report Builder	Drag-drop lab letterhead/footer customization	● Med	P2
Conditional PDF Formatting	Abnormal values auto bold+red in report	● High	P1

PART C

Developer Execution Plan

Weeks 1–16 → MVP Launch (Starting this weekend)



You start this weekend. Week 1 = Sprint 1. Every sprint = 1 week. 16 sprints = 4 months to MVP launch.

MONTH 1 — Foundation & Infrastructure (Weeks 1–4)

Week	Tasks	You / Delegate	Priority
W1	Monorepo setup (Turborepo) · GitHub repo · CI/CD pipeline (GitHub Actions) · Docker setup · ESLint + Prettier + Husky · AWS account setup · Dev/Staging environments	YOU	● BLOCKER
W1	PostgreSQL RDS setup (dev) · Redis ElastiCache setup · S3 bucket creation · Secrets Manager · Route53 + ACM SSL	YOU	● BLOCKER
W2	NestJS app skeleton · Multi-tenancy middleware (schema-per-tenant) · Tenant resolver (subdomain → schema) · Database migration system (TypeORM)	YOU	● BLOCKER
W2	Next.js 14 app setup · Auth pages (login, forgot password) · Design system setup (shadcn/ui tokens) · Storybook setup	YOU	● BLOCKER
W3	JWT auth with refresh tokens · Argon2id password hashing · Rate limiting (5 fails = 15min lockout) · RBAC guards (5 roles) · Audit log interceptor	YOU	Critical
W3	User management UI · Role assignment · Session management · Login/logout flow · JWT storage strategy (httpOnly cookie)	YOU	Critical
W4	Lab profile module · Sample master · Department master · Lookup tables (dropdowns) · Centre master (single centre) · Employee master	DELEGATE*	High
W4	BullMQ queue setup · Redis pub/sub for real-time · Socket.io setup · Service Worker (offline PWA skeleton) · IndexedDB setup	YOU	High

* Delegate = junior dev or freelancer can handle CRUD screens once DB schema and API patterns are established.

MONTH 2 — Core Modules (Weeks 5–8)

Week	Tasks	You / Delegate	Priority
W5	Test Master CRUD (API + UI) · Test groups + departments · Test parameters with normal ranges · Rate list per test · Compliment/discount rules	DELEGATE	● BLOCKER
W5	Bulk test import via CSV · Test catalog search (Redis cached) · Test profile/package bundles · Package discount logic	YOU	High
W6	Doctor Master CRUD · Commission % configuration · Compliment master · Doctor rate list · Centre master (multi) skeleton	DELEGATE	High
W6	Patient registration + UID generation · Patient search (full-text GIN index) · Patient history view · Demographics form	YOU	● BLOCKER
W7	Booking screen — CORE (most complex) · Patient autocomplete · Doctor selection (commission trigger) · Test selection with rates · Sample type + barcode generation · Payment (full/partial)	YOU	● BLOCKER
W7	Keyboard shortcuts (F1–F5, Ctrl+S, Ctrl+P) · Working slip print · Booking receipt print · Booking validation rules · Offline booking (IndexedDB)	YOU	● BLOCKER
W8	Booking financial logic · Discount application (manual + automatic) · Balance tracking · Cancel booking with audit remark · Balance receipt screen	YOU	● BLOCKER
W8	Barcode generation (Code128 + QR) · Barcode label print template · Sample tracking status · Barcode scanner integration (USB HID)	DELEGATE	High

MONTH 3 — Results, Reports & Delivery (Weeks 9–12)

Week	Tasks	You / Delegate	Priority
W9	Result entry form · Parameter-wise value input · Normal range validation · Abnormal flagging (auto bold+red) · Critical value flagging + urgent SMS · Auto-retest suggestion	YOU	● BLOCKER
W9	Workload dashboard · Real-time pending list (Socket.io) · TAT timer per test · Live TAT breach alert · Supervisor view · Technician view	YOU	● BLOCKER
W10	PDF engine setup (Puppeteer) · Pathology report template · Conditional formatting	YOU	● BLOCKER

Week	Tasks	You / Delegate	Priority
	(abnormal = bold+red) · QR code on report (anti-forgery) · Dynamic watermark (patient name + date) · Report preview in browser		
W10	Per-lab report template config · Lab letterhead + logo · Signature blocks (up to 5) · Footnotes · Barcode label template · Receipt/invoice template	DELEGATE	High
W11	MSG91 SMS integration · SMS templates (DLT registered) · Booking confirmation SMS · Result ready SMS with report link · Critical value urgent SMS	YOU	High
W11	WhatsApp Business API (Twilio) · PDF report delivery via WhatsApp · Report link delivery · Status tracking (sent/delivered/read) · AWS SES email with PDF attachment	YOU	 High
W12	Report Partly screen (selective print) · Internet Reporting screen · Email report to doctor/patient · Report status tracking · Report version history	DELEGATE	High
W12	Patient history screen (longitudinal) · All visits linked · Previous reports viewable · TAT analytics per department · Department-wise pending count	DELEGATE	High

MONTH 4 — MIS, Collections & Launch (Weeks 13–16)

Week	Tasks	You / Delegate	Priority
W13	Day Register report · Day Collection report (read replica) · Test Summary report · Balance Report · Cancel Booking report · Doctor-wise collection	DELEGATE	High
W13	Live collection counter (WebSocket) · Daily voucher (petty cash) · Export to Excel (ExcelJS) · Print all MIS reports · MIS date range filters	YOU	High
W14	Balance receipt module · Receipt cancellation with audit · Collection reconciliation · Doctor commission auto-calculation · Commission statement generation	DELEGATE	High
W14	DPDP Act 2023 — consent management · Data deletion request handler · Breach notification system · Aadhaar masking enforcement everywhere · Suspicious activity detection	YOU	High
W15	Full E2E test suite (Playwright) · API integration tests (Supertest) · Load testing (k6) — 200 concurrent bookings · Performance profiling · DB query optimization	YOU	Critical

Week	Tasks	You / Delegate	Priority
W15	Self-service onboarding wizard (5 steps) · Demo tenant with sample data · 14-day free trial flow · Billing/subscription setup (Razorpay) · Welcome email flow	YOU	Critical
W16	Launch checklist: SSL ✓ WAF ✓ Backups ✓ Monitoring ✓ Alerts ✓ · Production deployment (Blue-Green) · Smoke tests · First pilot lab onboarding · Go-live!	YOU	 LAUNCH
W16	Documentation: API docs (Swagger) · Admin guide · User manual (basic) · Video walkthrough (screen record) · Support channel setup	DELEGATE	High

 DELEGATION RULE	You (Full-Stack Dev): Architecture, auth, booking core, PDF engine, real-time, security, DevOps. Delegate: CRUD screens, report templates, documentation, basic MIS screens. Ratio: You do 60%, delegate 40%.
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Months 5–14

Phase 2 — Full Feature Parity (Months 5–8)

Month	Features	Effort	Why Critical
Month 5	Radiology Result Entry (USG/X-Ray) · Rich text editor (Quill.js) · Radiology report template	2 sprints	Revenue: radiology is 30–50% of lab income
Month 5	CMO Entry + PC-PNDT Register · Form F compliance · Monthly submission print · Legal inspection readiness	2 sprints	Legal compliance — mandatory for USG labs
Month 6	Doctor Billing Commission Reports · Period-wise statement · Paid/unpaid tracking · Commission settlement	1 sprint	Doctor retention — they won't refer if unpaid
Month 6	Multi-centre Support · Centre-wise collection · Centre-wise MIS · Cross-centre patient transfer · Centre admin role	2 sprints	Unlocks Business tier pricing (₹14,999/mo)
Month 7	Full MIS Panel (20+ reports) · Centre Account · Doctor Account · Out Source Lab reports · Previous collection · Voucher summary	3 sprints	Owner cannot manage without complete MIS
Month 7	Corporate Billing module · TPA/Insurance billing · Monthly consolidated invoice · Corporate client master	2 sprints	Unlocks corporate accounts = stable recurring revenue
Month 8	MFA (TOTP) for admin/owner · White-label (custom domain via CNAME) · IP allowlisting per tenant · Session concurrent control	2 sprints	Enterprise tier requirement
Month 8	Dynamic report builder (drag-drop) · Template marketplace (CBC, LFT, KFT pre-built) · Report versioning · Correction audit trail	2 sprints	Differentiation from competitors

Phase 3 — Enterprise & Scale (Months 9–14)

Month	Feature	Business Impact
Month 9–10	HL7/FHIR Integration · ABHA Health ID linkage · Hospital HIS connectivity	Unlocks hospital chain market
Month 9–10	LIS Instrument Auto-import · Analyzer result feed · Zero manual entry for instruments	Lab efficiency 10x — key selling point for big labs

Month	Feature	Business Impact
Month 11	AI abnormal result flagging · Smart patient search (fuzzy/phonetic) · TAT prediction ML model	First AI-powered lab software in India
Month 11	Mobile PWA (native feel) · Field collection app · Mobile result approval · Owner mobile dashboard	Collection staff efficiency
Month 12	Analytics Dashboard v2 · Revenue forecasting · Patient flow prediction · Predictive inventory	Owner strategic decisions
Month 13	Voice dictation for radiology (Whisper API) · Hindi + regional language support · OCR for paper reports	Radiologist delight — zero typing
Month 14	Franchise/chain hierarchy reporting · NABL/ISO QC module · API marketplace (3rd party) · SOC 2 Type II readiness	Lab chain deals — ₹50L+ contracts

Phase 1 Team (You + Delegates)

Role	Who	Responsibilities	Cost Estimate
Tech Lead / Architect	YOU	Architecture, auth, booking core, PDF, DevOps, security	Your time
Full-stack Dev (CRUD)	Freelancer	Master screens, basic MIS, report templates	₹25–40k/month
QA Engineer	Freelancer	Test automation, manual testing, regression	₹15–25k/month
UI/UX Designer	Freelancer	Design system, screen mockups, Figma	₹20–35k fixed
Healthcare Consultant	Part-time	Workflow validation, NABH/NABL compliance	₹10–15k fixed
DevOps (Part-time)	YOU	AWS setup, CI/CD, monitoring	Your time

 DELEGATE RULE	You ONLY do: Architecture decisions, complex business logic, security, performance, DevOps. Everything else (CRUD APIs, admin screens, report templates, documentation) should be delegated to save your time for high-value work.
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PART D

Business Analyst Execution Plan

NOTE

Since this is your company, YOU will also act as the BA for Phase 1. Consider hiring a part-time BA from Month 3 onwards as requirements grow complex.

BA Deliverables by Month

Month	BA Deliverable	Output Document
Month 1(Weeks 1–2)	Current state analysis · Stakeholder interviews · User persona definition · Pain point mapping	1. User Persona Document (5 roles)2. Current Workflow Diagrams3. Pain Points Register
Month 1(Weeks 3–4)	Requirements gathering · Feature backlog creation · User story writing · Acceptance criteria	4. Product Requirements Document (PRD)5. User Stories (Jira/Linear)6. Definition of Done checklist
Month 2(Weeks 5–8)	Wireframe review · Sprint planning support · UAT planning · Demo script creation	7. UAT Test Plan8. Sprint Demo Scripts9. Feature Flag Matrix
Month 3(Weeks 9–12)	UAT execution with pilot lab · Bug report triage · Change request management · Pilot feedback analysis	10. UAT Report11. Change Request Log12. Pilot Lab Feedback Summary
Month 4(Weeks 13–16)	Launch readiness checklist · User manual creation · Training material · Help documentation	13. Launch Readiness Report14. User Manual (role-wise)15. Video training scripts
Month 5+	Competitor analysis updates · Feature prioritization · NPS survey design · Customer interview analysis	16. Competitive Analysis Report (quarterly)17. Feature Priority Matrix18. Customer Insight Reports

User Personas (5 Primary Roles)

Persona	Primary Goal	Pain Points Today	Success Metric
Receptionist (Priya)	Register patients + collect payment fast	Manual entry, no keyboard shortcuts, slow software	< 2 min per booking
Lab Technician (Rahul)	Enter results accurately, fast	Switching screens, no normal range auto-check	< 5 min per result set
Lab Owner (Dr. Sharma)	See daily collection + business health	No mobile access, manual reports, no trends	Dashboard in < 30 sec
Accounts Staff	Reconcile collections +	Manual calculation, errors,	Zero reconciliation errors

Persona	Primary Goal	Pain Points Today	Success Metric
(Sunita)	doctor commissions	no audit trail	
Radiologist (Dr. Kumar)	Report findings quickly, professional output	Typing long reports, no template reuse	Report in < 10 min

Key User Stories — Phase 1 (Sample)

Format: As a [role], I want to [action] so that [outcome].

- As a Receptionist, I want to register a new patient in under 2 minutes using keyboard shortcuts so that I can handle peak morning rush without slowing down the queue.
- As a Lab Technician, I want the system to auto-highlight abnormal results in red so that I never miss flagging a critical value.
- As a Lab Owner, I want to see today's collection on my mobile phone at any time so that I can track business health while away from lab.
- As a Patient, I want to receive my report on WhatsApp as soon as it is ready so that I don't have to call the lab or visit again.
- As an Accounts Staff, I want doctor commission statements auto-generated for any date range so that I never have to manually calculate.
- As a Lab Owner, I want booking data to be saved offline during internet outage so that lab operations never stop.

Acceptance Criteria Template


EXAMPLE

Feature: Booking Screen✔ Patient can be registered in < 2 minutes✔ Test catalog search returns results in < 100ms✔ Barcode generates on save without extra click✔ Working slip prints automatically✔ Partial payment handled correctly (balance tracked)✔ Offline booking saves locally when internet down✔ Keyboard shortcut F1 opens new booking from any screen

Existing Competitors in Indian Lab Software Market

Software	Price Range	Cloud?	Mobile?	WhatsApp?	Weakness
Pathsys	₹15–40k/yr	Partial	No	No	Desktop-first, dated UI
SRL/Dr. Lal's (internal)	Not sold	Yes	Yes	No	Not available to smaller labs
eHospital/ HospitalRun	₹10–30k/yr	Partial	No	No	Complex, not lab-specific
Advik Path Lab	~₹20k one-time	No	No	No	Desktop only, no scale
PathCare Labs SaaS 	₹3k–15k/mo	 Full	 PWA	 Yes	(Yours — no weakness)

 **YOUR MOAT**

No competitor has: Offline-first PWA + WhatsApp report delivery + Keyboard shortcuts + QR anti-forgery reports + Self-service onboarding + Monthly SaaS pricing. You win on ALL these.

PART E

Marketing & Sales Execution Plan

Phase 1 GTM: Nail 10 Labs Before Scaling

**STRATEGY**

Do NOT try to sell to 100 labs in Month 1. Get 3–5 pilot labs, make them wildly successful, get video testimonials, then scale. Word-of-mouth in Indian lab community is extremely powerful.

Target Customer Segments

Segment	Profile	Size	How to Reach
Segment 1 (MVP)	Standalone lab, Tier-2/3 city, currently on desktop software	~50,000 labs in India	Facebook groups, Lab associations, WhatsApp groups
Segment 2 (Growth)	2–5 branch pathology chain, frustrated with coordination	~10,000 chains	Cold email, LinkedIn, direct sales calls
Segment 3 (Scale)	Hospital-attached lab wanting modern software	~5,000 hospitals	Hospital administrator LinkedIn + conferences
Segment 4 (Enterprise)	Diagnostic chain (10+ branches)	~500 chains	Direct enterprise sales, references

Pricing & Packaging

Tier	Price/Month	Target	What's Included
 Starter	₹2,999	Single lab, < 500 bookings/mo	Core booking, results, basic MIS, SMS, WhatsApp, QR reports
 Professional	₹6,999	Single lab, < 3,000/mo	All Starter + Radiology + Doctor billing + Full MIS + Custom reports + White-label
 Business	₹14,999	Multi-centre, < 10K/mo	All Professional + Multi-centre + Corporate billing + TPA + Priority support
 Enterprise	Custom	Lab chains, unlimited	All Business + HL7/FHIR + LIS integration + Dedicated server + SLA + Full white-label
 Free Trial	₹0 / 14 days	Any new lab	Full Professional features, no credit card required, sample data pre-loaded



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Offer 2 months free on annual payment: Starter ₹29,999/yr (save ₹6k), Professional ₹69,999/yr (save ₹14k). This improves cash flow and reduces churn.

Pre-Launch Phase (Months 1–3) — Build in Public

Month	Marketing Activities	Goal / Metric
Month 1	Set up social media (LinkedIn + Instagram + Facebook Page) · Share "Building a lab software in public" story · Join 5+ lab owner WhatsApp groups · Identify 3 pilot labs willing to beta test	50 LinkedIn followers · 3 pilot labs confirmed · 200 WhatsApp group members reached
Month 2	Post weekly product update videos (30 sec screen recordings) · LinkedIn articles: "Why Indian lab software is stuck in 2005" · Collect pain points from lab owners · Build email waitlist landing page	500 LinkedIn followers · 200 email waitlist signups · 10 lab owner interviews done
Month 3	Start pilot with 3 labs (free, in exchange for feedback + testimonial) · Document every feedback with photo/video · Create case study: "How Jerath Path Lab saved 2 hours/day" · Share progress on social media weekly	3 pilots running · 1 case study published · 1,000 LinkedIn followers · ₹0 revenue but strong word-of-mouth started

Launch Phase (Month 4) — Go Live

Week	Activity	Goal
W1 of M4	Product Hunt launch · LinkedIn announcement with product demo video (2 min) · Email waitlist announcement · Personal WhatsApp broadcast to all contacts	500 website visitors on launch day · 20 free trial signups
W2 of M4	Follow up with all trial signups (personal call/WhatsApp) · Share pilot lab testimonial video · Write "Launch Story" blog post · Announce in all lab WhatsApp groups	5 trials converted to paid · 50 total trials
W3–4 of M4	Cold email campaign to 500 Tier-2 city labs · Google Ads (test ₹5k budget) · "Lab software demo" Facebook Ad (₹3k) · Referral program launch: ₹1,000 per referral	10 paying customers · ₹30k MRR · Referral program live

Growth Phase (Months 5–8)

Month	Marketing Activities	Revenue Target
Month 5	SEO content: "Best lab software India 2025" · YouTube channel: Lab software tutorials (Hindi + English) · Attend 1 regional lab association event · Launch affiliate program for lab	₹1L MRR · 30+ paying labs

Month	Marketing Activities	Revenue Target
	equipment vendors	
Month 6	Partner with lab equipment suppliers (Sysmex, Mindray) for software bundling · Google Ads scaled to ₹30k/month · Case study: Multi-branch chain success story · LinkedIn ads targeting lab owners	₹2L MRR · 60+ labs
Month 7	Launch "PathCare Certified Lab" badge (marketing for their labs) · WhatsApp broadcast to 2,000+ lab contacts · PR: Pitch to HealthTech media (Inc42, YourStory, ET Health) · Channel partner program (IT vendors who service labs)	₹3.5L MRR · 100+ labs
Month 8	Enterprise outreach: 20 multi-branch chain direct meetings · Conference: NABL/NABH accreditation conferences · Launch enterprise tier with dedicated sales · NPS survey to existing customers, improve on feedback	₹5L MRR · 150+ labs · 2 Enterprise deals in pipeline

Scale Phase (Months 9–14)

Month	Activity	Revenue Target
Month 9	Dedicated sales rep hired · CRM setup (HubSpot) · Enterprise sales process formalized · Partnership with NABH consultants	₹8L MRR · 250+ labs
Month 10	Regional expansion push (South India market) · Tamil Nadu / Karnataka lab association events · Regional language UI (Tamil, Kannada, Telugu roadmap)	₹12L MRR · 350+ labs
Month 11–12	Franchise model: "PathCare Partner" for IT vendors to resell · Investor deck preparation if funding desired · Series A readiness (₹10L+ MRR is threshold)	₹18L MRR · 500+ labs
Month 13–14	1,000 lab milestone campaign · Press release + media coverage · Hospital chain enterprise deals · International expansion research (Bangladesh, Nepal, Middle East)	₹30L+ MRR · 1,000 labs

Sales Process for Lab Software

Stage	Action	Tool / Channel	Timeline
1. Prospecting	Find lab owner contacts via lab association lists, Google Maps, Justdial	Google Maps API, manual research, LinkedIn	Ongoing
2. Outreach	WhatsApp message: "Are you still using desktop software? See how 50 labs switched to cloud in 2 minutes."	WhatsApp Business · LinkedIn DM · Cold email	Day 1
3. Demo	15-min screen-share demo. Show booking → result → WhatsApp report in real-time. Let them feel the speed.	Google Meet / Zoom · Your live demo tenant	Day 3–5
4. Free Trial	Start 14-day trial with their actual test catalog pre-loaded (do this manually for first 20 customers)	Your platform · WhatsApp support	Day 5–7
5. Trial Support	Personal WhatsApp check-ins daily during trial. Solve every issue in < 1 hour.	WhatsApp · Screen-share support	Day 7–19
6. Close	"You've been using it for 14 days. Ready to continue? Annual plan = 2 months free."	WhatsApp · Razorpay payment link	Day 14–21
7. Referral	After 30 days: "Know any other lab owners? ₹1,000 credit for each referral who pays."	WhatsApp · Email	Day 30+

WhatsApp Sales Script (First Message)

 TEMPLAT E	Hi [Owner Name],Mainne dekha aap [City] mein [Lab Name] chalte hain — badiya!Kya aap abhi bhi desktop software use kar rahe hain reports ke liye?Humne ek cloud-based lab software banaya hai jismein:✅ Patient report WhatsApp pe automatically✅ Mobile pe daily collection dekhein✅ Internet na ho tab bhi kaam kare✅ 14 din free try kareinKya 10 minute mein demo dekh sakte hain? 🙏
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Content Marketing Plan

Platform	Content Type	Frequency	Goal
LinkedIn	Product updates · Founder story · Lab industry insights · Case studies	3x per week	B2B reach · Lab owners
Instagram / Facebook	Short video demos (30 sec) · Before/after comparisons · Customer stories	2x per week	Wider lab community
YouTube	Full tutorial videos (Hindi) · Feature walkthroughs · "Lab software tips" series	1x per week	SEO + Trust
WhatsApp Broadcast	Product updates · New features · Lab industry news · Tips	2x per month	Direct to lab owners
Blog / SEO	"Best lab software India" · NABH compliance guides · Lab business tips	2x per month	Organic traffic

The Big Picture: What Happens When

Month	 Development	 BA / Product	 Marketing	 Revenue
1	Foundation, Auth, CI/CD, DB setup	User research, PRD, user stories	Social setup, waitlist page, join lab groups	₹0 — building
2	Test catalog, Doctor/Patient masters, Booking core	Wireframe review, UAT planning	Weekly product videos, pain point content	₹0 — pilot prep
3	Results, PDF engine, WhatsApp delivery	Pilot lab UAT, feedback collection	3 pilot labs live, case study writing	₹0 — pilot (free)
4	MIS, collections, QA, launch	Launch readiness, user manual	Product Hunt, LinkedIn launch, trial campaign	₹30–50k MRR
5–6	Radiology, CMO/PC-PNDT, Doctor billing	Phase 2 requirements, competitor analysis	SEO content, YouTube, equipment partnerships	₹1–2L MRR
7–8	Multi-centre, Full MIS, Corporate billing, MFA	Enterprise requirements, NPS survey	PR outreach, lab conferences, affiliate program	₹3–5L MRR
9–10	HL7/FHIR, LIS instrument integration	Enterprise onboarding docs, SLA docs	Dedicated sales rep, CRM setup, enterprise outreach	₹8–12L MRR
11–12	AI features, mobile PWA, analytics v2	AI feature specs, mobile UX review	1,000 lab milestone, Series A prep if needed	₹18–25L MRR
13–14	Voice dictation, franchise hierarchy, NABL module	International market research	International expansion research, hospital chain deals	₹30L+ MRR

 YOUR NORTH STAR METRICS	Month 4: 10 paying labs · Month 8: 150 paying labs · Month 12: 500 paying labs · Month 14: 1,000 paying labs · ARR: ₹60Cr at 5,000 labs (₹10k avg ARPU)
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APPENDIX

Folder Structures, DB Schema Reference & API Design

A. Frontend Folder Structure (Next.js 14)

```
src/├─ app/                                # Next.js App Router|├─ (auth)/                                # Login, forgot password|├─ (dashboard)/                          # Booking screen (F1)|├─ results/                             # Result entry (F2)|├─ workload (F4)|├─ mis/                                # MIS reports|├─ settings/                            # Lab config|├─ api/                                # BFF API routes|├─ layout.tsx├─ components/├─ ui/                                # shadcn/ui base|├─ forms/├─ tables/                            # Virtual scrolling data grids|├─ reports/                            # PDF preview components|├─ layout/                            # App shell, keyboard shortcuts├─ lib/├─ api/                            # TanStack Query hooks|├─ auth/├─ offline/                            # IndexedDB, sync queue|├─ permissions/├─ store/                            # Zustand stores|├─ authStore.ts|├─ offlineStore.ts                    # Offline queue state|├─ uiStore.ts├─ styles/├─ globals.css└─ design-tokens.css                # All CSS variables
```

B. Backend Folder Structure (NestJS)

```
src/├─ modules/├─ auth/                            # JWT, MFA, Passport strategies|├─ tenant/                            # Schema-per-tenant middleware|├─ booking/                            # Core transaction domain|├─ patient/                            # Patient domain|├─ test-catalog/                        # Test master, parameters, groups|├─ results/                            # Result entry, abnormal flagging|├─ billing/                            # Receipts, commissions|├─ doctor/                            # Provider domain|├─ radiology/                          # USG/X-Ray reports, CMO|├─ mis/                                # All MIS report queries|├─ notifications/                      # SMS, WhatsApp, Email, Push|├─ reports/                            # PDF generation (Puppeteer)|├─ admin/                              # System admin, backups|├─ shared/├─ database/                          # TypeORM config, migrations|├─ audit/                            # Immutable audit log interceptor|├─ cache/                            # Redis service wrapper|├─ storage/                          # S3 service wrapper|├─ queue/                            # BullMQ job definitions└─ main.ts
```

C. Key API Endpoints Reference

Method	Endpoint	Auth	Description
POST	/api/v1/auth/login	Public	Tenant-aware login, returns JWT
POST	/api/v1/bookings	Receptionist+	Create new booking with tests
GET	/api/v1/bookings/workload	All staff	Real-time pending workload list
POST	/api/v1/results/:bookingId	Lab Tech+	Enter test result values
GET	/api/v1/reports/:bookingId/pdf	All staff	Generate + return PDF URL
POST	/api/v1/reports/:id/deliver	Lab Tech+	Send WhatsApp + email + SMS
GET	/api/v1/mis/day-collection	Admin+	Day collection summary (read replica)
GET	/api/v1/patients/:uid/history	All staff	Full longitudinal patient history
POST	/api/v1/receipts	Cashier+	Collect balance payment
GET	/api/v1/tests/search?q=	All staff	Redis-cached test catalog search

D. Data Retention & Compliance Policy

Data Age	Storage Location	Access Speed
0–1 year (Hot)	Primary RDS PostgreSQL	Full speed, all queries
1–3 years (Warm)	Primary RDS (compressed partitions)	Normal speed
3–7 years (Cold)	S3 Glacier (pg_dump per tenant)	Restore on request (~4hrs)
7+ years (Archive)	S3 Deep Archive	Legal compliance copy only
PC-PNDT Register	Minimum 2 years hot	Immediate (regulatory inspection)
NABL/NABH Records	Minimum 5 years	Per accreditation requirement

E. Weekend #1 Checklist — Start This Weekend

 **THIS WEEKEND**

You said you start this weekend. Here is exactly what to do in the first 2 days to set yourself up for success.

Saturday (Day 1):

- 1. Create GitHub organization + monorepo (Turborepo) — 30 min
- 2. Set up Next.js 14 app with TypeScript + ESLint + Prettier — 45 min
- 3. Set up NestJS app with TypeScript — 30 min
- 4. Configure shared tsconfig and types package — 20 min
- 5. Set up GitHub Actions CI (lint + typecheck on PR) — 45 min
- 6. Create AWS account (if not done) + set up IAM user — 30 min
- 7. Create dev PostgreSQL RDS (db.t3.micro) + Redis ElastiCache — 1 hr
- 8. Set up Docker + docker-compose for local dev — 30 min
- 9. Create .env.example with all required environment variables — 20 min

Sunday (Day 2):

- 10. Design and implement tenant schema (public + tenant_{code}) — 2 hrs
- 11. Write first 3 TypeORM migrations (lab_profile, departments, samples) — 1 hr
- 12. Implement tenant resolver middleware (subdomain → schema) — 1 hr
- 13. Set up Storybook — 30 min
- 14. Create design tokens CSS file (colors, spacing, typography) — 30 min
- 15. Set up shadcn/ui — 20 min
- 16. Deploy "Hello World" to AWS (ECS or even just EC2 for now) — 1 hr
- 17. Create project board (Linear or GitHub Projects) with all Sprint 1 tasks — 30 min

End of Weekend Goal: Monorepo running locally, CI green, DB connected to multi-tenant schema, deployed to AWS dev environment. You're officially building.

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